

GRANITE FALLS, MN, USA

WMO: 722033

Lat: 44.753N

Lon: 95.556W

Elev: 1047

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-9.1	-19.8	1.9	-13.1	-15.4	2.5	-8.8	32.1	11.6	28.5	14.3	6.9	260	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	90.2	73.9	87.2	72.4	84.4	70.5	77.5	86.8	75.3	83.6	73.4	81.1	12.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	134.8	83.7	72.6	125.7	81.3	70.7	117.7	79.0	41.7	87.0	39.5	84.0	37.6	81.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	24.3	20.8	DB	-22.5	95.8	5.8	2.4	-26.6	97.5	-30.0	98.9	-33.3	100.2	-37.5	101.9	(4)
(5)			WB	-22.8	81.1	5.7	2.0	-26.9	82.5	-30.2	83.6	-33.4	84.7	-37.5	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	15.0	17.0	31.7	45.1	57.7	68.7	72.3	69.0	62.3	47.7	33.8	19.1		
	DBStd	22.79	13.74	12.69	12.91	10.42	9.14	6.18	5.56	5.44	8.32	9.41	11.10	12.31		
	HDD50	4450	1086	925	582	209	32	0	0	0	7	156	495	958		
	HDD65	7900	1551	1345	1034	599	261	30	9	23	148	541	936	1423		
	CDD50	2657	0	0	13	63	272	560	690	589	376	85	9	0		
	CDD65	634	0	0	1	3	36	140	234	147	67	6	0	0		
	CDH74	6301	0	0	6	53	502	1389	2336	1258	672	84	1	0		
	CDH80	1955	0	0	1	9	181	449	781	324	191	19	0	0		
	WSAvg	9.5	10.2	10.5	10.5	11.4	11.4	9.1	7.0	6.4	8.0	9.4	10.1	9.7		
	PrecAvg	27.7	0.7	0.7	1.6	2.6	3.3	4.3	3.5	3.5	3.1	2.5	1.3	0.8		
(15) Precipitation	PrecMax	39.9	2.1	1.5	4.1	6.5	6.4	9.8	7.0	7.9	8.3	7.1	4.3	2.7		
	PrecMin	21.4	0.0	0.2	0.2	0.6	0.9	1.9	0.7	0.6	0.5	0.4	0.0	0.0		
	PrecStd	5.5	0.6	0.4	0.9	1.4	1.5	2.0	1.6	1.7	1.9	1.7	1.3	0.7		
	PrecStd	5.5	0.6	0.4	0.9	1.4	1.5	2.0	1.6	1.7	1.9	1.7	1.3	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	54.5	71.6	80.3	92.4	93.8	93.3	90.1	90.2	82.4	68.1	49.7		
		MCWB	39.0	49.7	57.0	60.1	64.9	72.8	78.6	76.8	72.3	62.4	54.5	43.8		
	2%	DB	38.4	44.2	62.8	73.3	84.9	88.7	90.1	86.7	85.4	74.8	61.0	42.2		
		MCWB	35.4	39.8	54.2	56.7	63.5	71.0	76.7	74.6	70.1	59.8	50.9	37.6		
	5%	DB	35.9	38.1	55.1	67.9	79.8	85.4	87.1	83.7	81.6	68.6	54.6	37.4		
		MCWB	33.8	35.0	49.8	53.2	61.8	69.7	74.1	72.5	68.0	56.9	46.7	35.1		
	10%	DB	32.8	34.4	49.1	62.9	74.2	81.7	84.2	81.1	77.5	63.4	49.4	34.6		
		MCWB	31.6	32.3	44.4	51.3	59.4	67.1	71.9	70.0	65.9	53.8	43.4	32.8		
	0.4%	WB	39.7	49.9	60.3	63.1	70.6	77.5	81.5	78.7	76.3	66.9	57.7	45.2		
		MCDB	44.8	54.8	67.0	73.2	83.4	88.5	90.6	87.2	84.5	75.7	66.8	48.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	40.0	54.8	59.5	66.9	74.7	78.5	76.5	73.3	62.0	51.7	38.1		
		MCDB	38.1	43.5	60.3	70.6	78.4	84.3	87.3	84.6	80.5	71.1	58.7	41.3		
	5%	WB	33.9	35.5	49.8	55.9	64.2	71.7	76.2	74.2	70.9	58.4	47.6	35.3		
		MCDB	35.7	38.0	56.2	64.3	75.5	81.5	84.7	81.3	78.7	66.9	53.7	37.5		
	10%	WB	31.8	32.5	44.6	52.7	61.7	69.5	74.0	72.1	68.2	54.5	43.6	33.1		
		MCDB	32.9	34.3	49.0	61.4	71.0	78.7	82.1	78.8	75.2	61.8	49.5	34.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	18.1	17.0	20.4	21.8	20.2	21.5	20.9	22.9	20.8	18.4	16.0		
		MCDBR	18.8	20.7	26.2	31.0	30.5	24.9	22.6	23.2	26.8	29.1	27.3	18.2		
		MCWBR	15.9	16.4	17.2	17.3	13.7	11.2	11.7	12.3	13.2	16.2	17.4	15.1		
		MCDBR	17.3	20.0	25.2	24.7	24.0	21.2	20.9	20.4	22.2	25.0	25.1	17.7		
	5% WB	MCWBR	14.6	15.9	17.0	15.4	12.8	11.5	12.0	11.8	12.8	15.5	16.9	14.9		
		MCWBR	14.6	15.9	17.0	15.4	12.8	11.5	12.0	11.8	12.8	15.5	16.9	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.275	0.307	0.360	0.381	0.394	0.401	0.407	0.365	0.320	0.282	0.262		
	taud		2.458	2.430	2.446	2.342	2.346	2.368	2.334	2.295	2.391	2.509	2.533	2.468		
	Ebn at Noon		278	293	296	286	282	278	274	268	271	272	268	266		
	Edh at Noon		22	27	31	37	38	38	39	39	32	25	20	20		
(44) All-Sky Solar Radiation	RadAvg		580	912	1211	1502	1728	1925	2080	1716	1337	870	568	425		
	RadStd		65	106	120	146	128	132	70	140	99	116	50	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page